

AMENDMENT NO. 3 MAY 2009
TO
IS 1038 : 1983 SPECIFICATION FOR STEEL
DOORS, WINDOWS AND VENTILATORS

(*Third Revision*)

[Page 4, clause **0.6** (see also Amendment No. 1)] — Delete the last two sentences.

(Pages 22 and 23, clauses **6.1.1** to **6.1.1.3**) — Substitute the following for the existing clauses:

6.1.1 Requirements of Welded Joints

6.1.1.1 Visual inspection test — When the corners of the frames are visually inspected after removing the paint, the joints shall conform to the following:

- a) Welds shall have been made all along the place of the meeting the members,
- b) Welds shall have been properly ground, and
- c) The cross sections of the corners shall show a completely solid joint with no visible cavities.

6.1.1.2 Macro-examinations — Two opposite corners of the frames shall be cut and prepared for macro examination according to IS 3600 (Part 9) : 1985** and examined with hand lens of magnification of not less than 5. Complete cross section of the corner shall show a joint completely solid and with no visible cavities, under-cutting and overlaps.

6.1.1.3 Micro-examination — From the two opposite corners obtained for macro-examination, one corner shall be subjected to micro examination at three places: (i) at the centre of the outside flange, (ii) at the centre of the inside flange, and (iii) at the centre of web. For this, the flanges and the web of section shall be cut with the help of a saw, the cut surface shall be polished and etched according to IS 7739 (Part 1) : 1975†† and examined at the weld at a magnification of not less than 100 to check proper fusion all along the thickness of the section in accordance with IS 3600 (Part 9) : 1985**. This examination shall also reveal the absence of cavities, cracks, porosity, entrapped slag, penetration defects and fusion defects.

6.1.1.4 Fillet Weld Test — The remaining corner obtained for macro-examination shall be subjected to fillet weld test. The outside flange and inside flange shall be removed by saw cutting and remaining fillet shall be fractured by hammering. The fractured surface shall be free from slag inclusion, porosity, cracks, penetration defects and fusion defects.

(Page 22, footnotes) — Add the following footnotes at the end:

***Method of testing fusion welded joints and weld metal in steel : Part 9 Macro and micro examination (*second revision*).

††Code of practice for preparation of metallographic specimens : Part 1 General features.’

[Page 23, clause **6.1.2**, last sentence (see also Amendment No. 2)] — Substitute the following for the existing sentence:

‘All sub-dividing and glazing bars shall be tenoned and riveted into the frames by tenon-riveting machine or any other suitable equipment. No welding shall be permissible in the entire joint including the meeting place of the glazing and subdividing bars as also the rivet-head. The rivet-head shall be free from any weld or slag deposit and should have been formed by riveting of the projection made out of the subdividing/glazing bar itself.’

(Page 23, clause **6.2**, first para, last sentence) — Substitute the following for the existing:

‘The hinge pin shall be of galvanized steel and the washer shall of brass or stainless steel, of suitable thickness.’

[Page 23, clause **6.2.1.1**, last sentence (see also Amendment No. 2)] — Substitute the following for the existing:

‘The hinge pin of diameter not less than 6 mm shall be of galvanized steel and washers shall be of brass or stainless steel of suitable thickness.’

(Page 23, clause 6.2.2, last sentence) — Substitute the following for the existing:

‘The handle plate with wall thickness of not less than 3 mm shall be riveted to the opening frame.’

(Page 29, clause 6.2.2.3) — Add the following matter at the end of the clause:

‘The fixing screw, shake proof washer, plain washer and plain washer fixed with the handle shall be of galvanized steel.’

[Page 29, clause 6.2.3, first sentence (see also Amendment No. 2)] — Add the following matter at the end of the clause:

‘Steel peg, pin and locking bracket with wall thickness of not less than 3 mm, shall be riveted to the window.’

(Page 37, clause 8.1) — Substitute the following for the existing clause:

‘8.1 Pretreatment

All the steel surfaces shall be thoroughly cleaned free of rust, millscale, dirt, oil, etc, by mechanical means, like sand or shot blasting or by chemical means, like pickling and then finished either with phosphating and painting (see 8.1.1); or by hot dip galvanizing (see 8.1.2), as may be agreed to between the purchaser and the manufacturer.’

[Page 37, clause 8.1.1 (see also Amendment No. 1)] — Delete the clause and renumber subsequent clauses.

[Page 39, clause 8.1.2 (see also Amendment No. 1)] — Substitute the following for the existing clause:

‘*Phosphating and Painting* — After pretreatment of the surfaces, the units shall be dipped in phosphating solution in accordance with the requirements laid down in IS 1477 (Part 1) : 2000*. This shall be followed by one coat of paint by dipping the complete unit in bath of a suitable primer, such as red oxide zinc chrome primer (see IS 2074 : 1992† and then air drying.’

(Page 39, clause 8.1.3) — Substitute the following for the existing clause:

‘*Hot Dipped Galvanizing* — After pretreatment of the surface, the units shall be dipped in a bath of molten zinc in accordance with the requirements laid down in IS 1477 (Part 1) : 1971**. The thickness of coating shall be uniform and not less than 0.5 kg/m².’

(Page 39, footnote) — Substitute the following for the existing footnote:

*Code of practice for painting of ferrous metals in buildings: Part 1 Pretreatment (second revision).

†Ready mixed paint, air drying, red oxide zinc chrome, priming — Specification (second revision).’